

XR for educational anatomy: an Immersive human atlas with chatbot

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ABSTRACT

Immersive reality has emerged as a groundbreaking and promising approach for teaching anatomy, providing unique learning experiences while complementing traditional educational methods. Existing immersive software, such as Organon® and HoloLens®, offer valuable tools but remain limited in anatomical detail and interactivity. We present a novel educational approach: an immersive human atlas that integrates anatomical slices from the Visible Korean Human dataset (male and female) into a 3D environment. This platform enhances anatomical visualization and incorporates a Large Language Model (LLM) to support user interaction and understanding. By combining precise anatomical slices within the 3D model, the atlas provides an enriched and comprehensive learning experience for medical education.

Index Terms: Extended Reality, Immersive Environments, Anatomy Education, Large Language Models, Cross Computing

1 INTRODUCTION

Teaching human anatomy is fundamental for medical professionals, particularly radiologists and surgeons. Traditionally, cadaver dissection has been the cornerstone of anatomy education [7, 9, 10]. However, this approach faces multiple challenges, including the increasing number of students, ethical concerns, and the scarcity of cadavers. These limitations have prompted a transition towards innovative teaching methods [12, 15] that leverage digital tools, particularly immersive technologies such as virtual reality (VR). Various anatomy learning software is now available on computers, tablets, and smartphones. Despite their accessibility, these tools often exhibit quality and anatomical accuracy inconsistencies. Furthermore, few applications integrate immersive virtual reality environments, limiting their ability to deliver an optimal educational experience [14].

The use of immersive VR in anatomy education offers substantial advantages compared to traditional teaching methods. Recent advancements in immersive anatomy education have demonstrated the potential of virtual reality to enhance spatial understanding and interactivity [3, 5]. These advantages can be summarized as follows:

Improved Spatial Awareness and 3D Understanding

VR enables students to explore three-dimensional anatomical structures, enhancing their spatial awareness and comprehension of body relationships. Rotating, zooming, and interacting with virtual models provides an unparalleled view of anatomical dynamics, offering a richer learning experience [17].

Increased Interactivity and Engagement Immersive VR environments foster active participation. Students can manipulate virtual anatomical structures, encouraging curiosity and deeper engagement with the content. This interactivity transforms learners from passive observers into active participants.

Improved Knowledge Retention and Scores Evidence suggests that VR significantly improves knowledge retention and learning outcomes in anatomy education compared to traditional methods. In particular, sessions lasting 30 minutes or more have been shown to yield greater knowledge gains due to the immersive experience [14].

Personalized Learning Experience VR provides a tailored educational environment, allowing students to learn independently. Customizable features enable independent exploration of the human body, addressing individual learning needs and facilitating in-depth understanding.

In this context, we propose a novel approach: an immersive human atlas that integrates anatomical slices from the Visible Korean Human dataset into a 3D virtual environment. This platform combines high-resolution anatomical data with interactive tools, offering students a dynamic and comprehensive learning experience.

2 METHODS

We constructed the 3D anatomical model through the meticulous manual segmentation of 8,000 cryosection images from the Visible Korean Human datasets, including both male [12, 13] and female [11]. We performed this process using *Winsurf* [15], which enabled us to contour each anatomical structure with high precision. We obtained the cryosection images using a dedicated cryomicrotome, ensuring ultra-high-resolution and anatomically accurate cross-sectional visualization.

After segmentation, we imported the resulting mesh files into *Blender*¹, a 3D modeling software widely used for scientific visualization and XR development. Within Blender, we applied extensive refinements to optimize the anatomical models. These refinements included mesh decimation to efficiently reduce polygons, ensuring compatibility with VR systems without compromising visual quality. We used surface smoothing techniques to eliminate segmentation artifacts, while subdivision modeling preserved fine structural details critical for anatomical fidelity. This process ensured that the final 3D models were visually accurate and computationally optimized for rendering in VR environments.

We segmented 1,200 anatomical structures to create a comprehensive whole-body representation encompassing skin, muscles, tendons, ligaments, nerves, arteries, veins, and organs. We organized these structures into complete physiological systems, including the integumentary, muscular, skeletal, nervous, cardiovascular, respiratory, gastrointestinal, urinary, reproductive, endocrine, and lymphatic systems, thus achieving an extensive and detailed virtual anatomical atlas.

¹Blender. <https://www.blender.org>. Accessed: 2024-12-22.

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We imported the refined 3D models into Unity², a leading real-time development engine for XR applications. We used the platform as the foundation to integrate immersive functionalities tailored for deployment on the Oculus Quest 2 VR headset. By leveraging Unity's XR Interaction Toolkit and associated VR software development kits (SDKs), we implemented advanced interaction mechanisms, including spatial tracking for precise controller-based manipulation and haptic feedback, to enhance the realism of the user experience. Real-time rendering and optimization strategies ensured the VR application ran smoothly on standalone hardware while maintaining high-fidelity anatomical detail.

The resulting system allowed for dynamic visualization and interaction with anatomical models. Users could display the entire human body or isolate individual structures and systems according to their educational goals, enhancing the platform's utility for interactive and personalized anatomy education.

3 RESULTS

The developed user interface provides a natural and immersive platform for interacting with the 3D anatomical model. Core functionalities include spatial navigation and 3D manipulation within the virtual environment, enabling users to explore the human body from multiple perspectives. These features are seamlessly accessible through an interactive menu panel adjacent to the anatomical model, as shown in Figure 1 which illustrates the reconstructed 3D mesh model, highlighting skeletal and muscular systems. The menu panel is divided into two functional sections: the left side controls body manipulation tools such as rotation, translation, and scaling, while the right side allows users to selectively display or hide entire anatomical systems, including muscles, bones, nerves, vessels, and organs. This clear layout ensures efficient interaction, supporting a detailed and systematic understanding of anatomical structures.

In addition to full-system visualization, the platform allows users to isolate specific anatomical regions, such as the head and neck, abdomen, pelvis, lower limb, and upper limb. This feature, illustrated in Figure 2, facilitates a focused study of regional anatomy, offering flexibility for learners and educators to tailor their exploration to specific areas of interest.

To complement its visualization features, the platform integrates an LLM, which provides detailed descriptions of selected anatomical structures. As shown in Figure 3, a dedicated menu panel delivers anatomical definitions in three languages, with an option for voice output. This multilingual functionality enhances accessibility for global learners, while the voice command control at the bottom-right corner allows users to switch languages seamlessly. A unique tool—the “hide bones” feature—further enhances interactivity by rendering bones transparent, providing unobstructed views of deeper structures, such as the brain through the cranial cavity.

A key innovation of the platform is its ability to integrate original axial anatomical slices directly into the 3D model. This feature bridges the gap between traditional cross-sectional imaging and virtual anatomy, allowing learners to correlate real anatomical slices with their corresponding three-dimensional structures. Figure 4 demonstrates the integration of an axial slice at the L3 vertebral level, where bones, vessels, and nerves are distinctly visualized within the 3D mesh. This functionality enhances spatial understanding and reinforces the connections between 2D sectional imaging, such as CT or MRI, and 3D virtual representations.

The platform delivers a natural and interactive learning environment that combines real-time 3D manipulation, system-based and region-specific exploration, multilingual anatomical definitions, and the integration of axial anatomical slices. The system offers a comprehensive, dynamic, learner-centered approach to anatomy education by uniting high-resolution anatomical models with advanced virtual reality tools.

²Unity. <https://unity.com/> Accessed: 2024-12-22.

4 CHATBOT INTEGRATION USING LLM

Our project integrates a chatbot powered by a Large Language Model (LLM) from OpenAI into DIVA VR to provide detailed anatomical descriptions. The chatbot was designed to deliver consistent and accurate content, offering comprehensive explanations of anatomical structures. This feature enhanced the interactive experience by providing users with clear and informative guidance within the virtual environment.

5 VIRTUAL DISSECTION USING TABLETS AND VR

Virtual dissection represents a transformative alternative to traditional cadaveric dissection, leveraging augmented reality (AR) and virtual reality (VR) to offer interactive learning experiences. Platforms such as Anatomy Studio have demonstrated how AR and VR can facilitate collaborative dissection through augmented 3D reconstructions [6]. These systems allow users to precisely manipulate anatomical models, enhancing spatial understanding and knowledge retention [16, 18], via a minimal calligraphic interface [1].

Using tablets and standalone VR headsets provides flexibility, enabling learners to engage in both independent and group-based explorations. Interactive features such as stereoscopic visualization, adjustable transparency, and dynamic slicing allow users to explore anatomical structures layer by layer [2, 4].

Figure 5 illustrates the user interface of a virtual dissection application, showcasing tools such as point distance adjustment, line thickness control, and color selection. These tools enhance precision when annotating anatomical features, facilitating an interactive experience.

Virtual dissection also fosters a multi-sensory learning experience, integrating visual, auditory, and tactile feedback to enhance comprehension. Studies have shown that immersive technologies evoke strong emotional engagement, leading to higher knowledge retention compared to traditional methods [6, 14]. Virtual dissection platforms are improving anatomy education by combining real-time 3D models with tools for customization, annotation, and multilingual AI assistance.

As shown in Figure 6, the platform supports stylus input, allowing users to annotate anatomical features directly on a tablet device. This feature promotes accessibility and precise interaction, enabling detailed exploration of anatomical datasets.

A cross-reality application developed for this purpose provides a versatile and highly interactive platform. It supports desktop and Android devices and is built using Unity, enabling compatibility with many systems. The application includes a navigable image dataset derived from the Visible Human Female project, enabling users to draw up to 256 contours per slice for each anatomical structure. Spline interpolation achieves volume reconstruction, resulting in highly accurate 3D models.

The platform supports exporting anatomical structures in standard formats such as STL and OBJ, making it compatible with various 3D modeling tools and printers. Additionally, it incorporates advanced remote collaboration features, allowing users to share contours, real-time drawing inputs, and voice communication. These capabilities are powered by Photon's PUN2 networking platform, with authentication handled via Google's Firebase.

The application optimizes performance by using a tiling technique for its image dataset. The original images are rescaled into four resolutions (4096, 2048, 1024, and 512 pixels) and divided into 256 by 256-pixel tiles in PNG format. These tiles are optimized through compression and palette reduction, ensuring efficient memory usage. Only the tiles required for a given zoom level are loaded based on user input, minimizing computational demands and enhancing responsiveness.

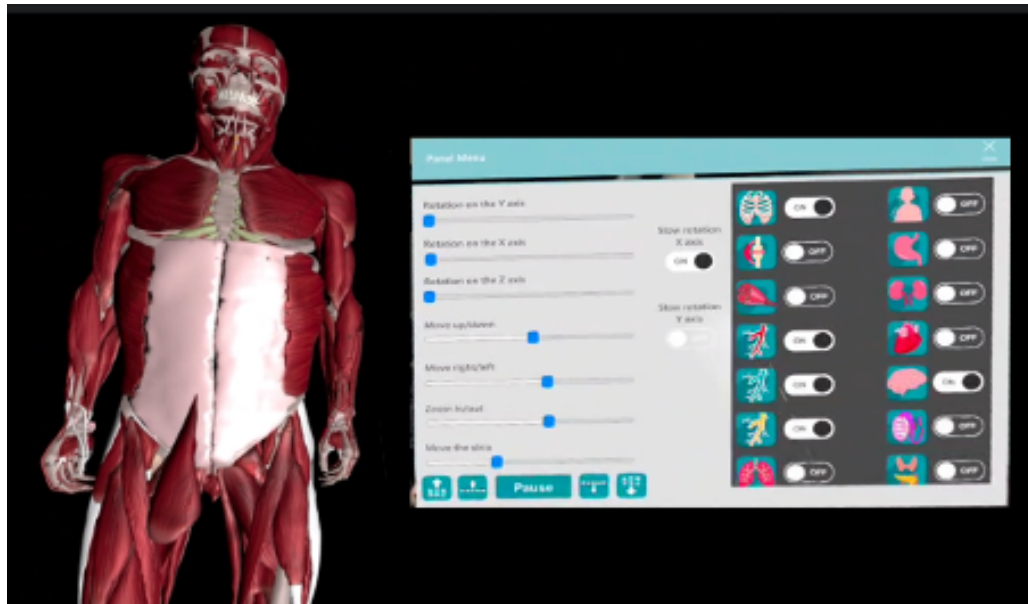


Figure 1: Reconstructed 3D mesh model with bones and muscles with the menu panel. The left part controls body manipulation, and the right part displays the systems.

6 SYSTEM ARCHITECTURE

The immersive anatomy education platform is built around a cross-reality application that integrates key components to deliver a interactive learning experience. At the system's core is a 3D Unity-based framework that handles the visualization and interaction with anatomical models and VR headset integration. Unity ensures high-fidelity visuals and smooth interactions for an immersive experience.

The platform relies on an input pipeline that processes anatomical meshes created using Blender from hand-digitized anatomical structures. These meshes are stored in a database and imported into Unity for rendering and user interaction, ensuring accurate representation of anatomical details. A chatbot powered by a Large Language Model (LLM) is also integrated into the application to provide detailed anatomical explanations. The chatbot supports multiple languages and offers voice output, making the platform accessible to a diverse audience, connecting to the cross-reality application via API, and delivering context-aware responses to enhance user engagement.

The architecture connects these components through efficient data flows, where the input pipeline supplies anatomical meshes to Unity, and the chatbot delivers interactive explanations to users. This integration bridges the gap between traditional anatomy education and modern immersive tools, creating a dynamic and effective educational experience.

7 PRELIMINARY EVALUATION

To assess the potential impact and usability of the immersive anatomy education platform, a preliminary evaluation was conducted with a small group of participants, including medical students, educators, and healthcare professionals. The evaluation focused on three key areas: usability, educational effectiveness, and user engagement.

Usability: Participants reported that the interface was easy to navigate, with features like anatomical mesh visualization and chatbot

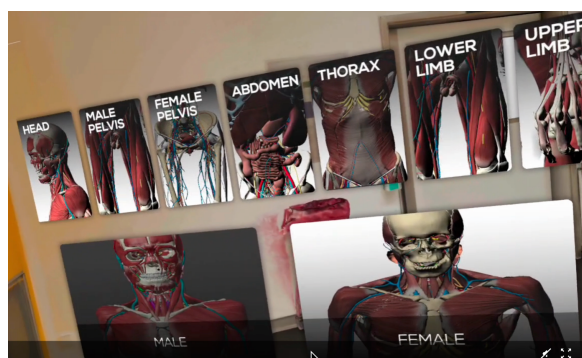


Figure 2: The menu choice for the whole-body reconstruction (male & female) and the regions.

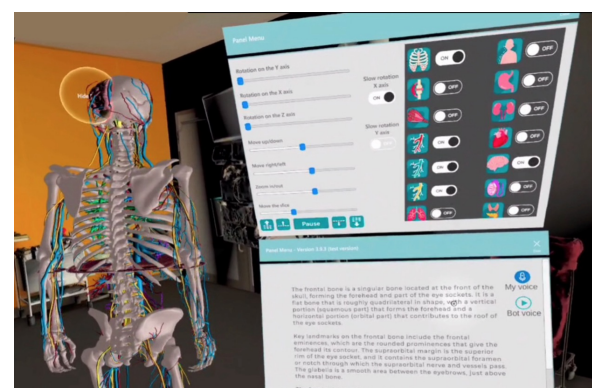


Figure 3: On the right, the menu panel (up) and the description of the selected element with the chatbot (frontal bone). On the left, the 3D reconstructed body with slices and the "hide bones" tool to see the brain through the cranium.

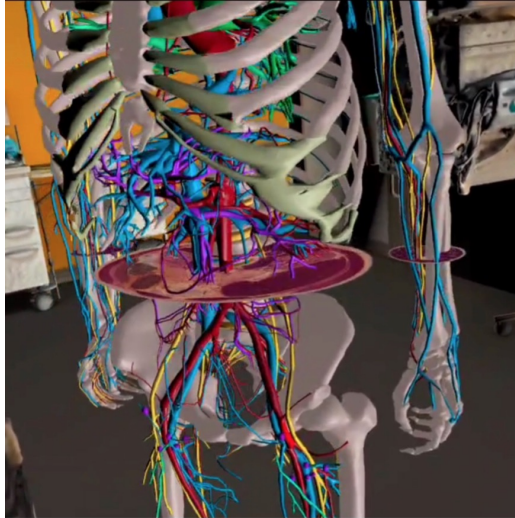


Figure 4: The display of the anatomical slice at the level of L3 inside the 3D mesh model with bones, vessels, and nerves.

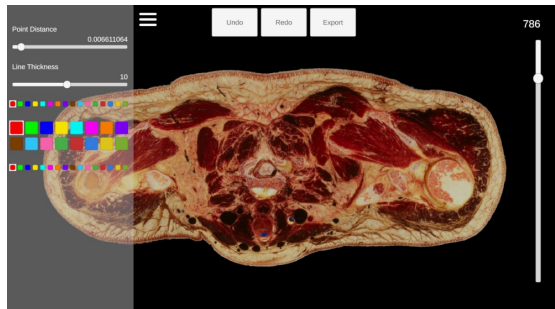


Figure 5: User interface showing tools for point distance, line thickness, and color selection, enabling precise annotation and visualization of anatomical structures.



Figure 6: Using a stylus to annotate anatomical features on a tablet device, illustrating the platform's interactivity and accessibility.

interaction. The VR functionality, in particular, was praised for its ability to create an immersive learning environment. Minor suggestions included improving the menu responsiveness and adding more customizable tools.

Educational Effectiveness: The participants found the interactive 3D models and detailed anatomical explanations highly informative. The chatbot integration provided context-aware anatomical

descriptions, enabling a deeper understanding of complex structures. Several participants indicated that the platform helped bridge the gap between textbook learning and real-world anatomy practice, although they preferred a speech-based interaction.

User Engagement: The cross-reality application was engaging. Features such as regional dissection and the ability to collaborate remotely were cited as key contributors to interest. The system's flexibility to support different devices, including tablets and VR headsets, was a major advantage.

While the feedback was mostly positive, participants suggested enhancements, such as integrating haptic feedback and speech interfaces for a more tactile experience and expanding the anatomical database to cover additional body systems. These insights will guide future iterations of the platform.

The preliminary evaluation indicates that the platform holds significant promise as an educational tool for anatomy, combining cutting-edge technology with practical utility to transform traditional learning approaches.

8 LIMITATIONS

While the platform offers promising features, it faces several limitations due to its preliminary nature. Integrating key components—such as the 3D Unity-based visualization, anatomical mesh pipeline, and chatbot—remains partial, leading to data flow and communication inefficiencies. Additionally, the lack of a speech-based interface restricts interactivity, and the system currently supports only basic 3D navigation, lacking advanced interaction techniques like gesture recognition or dynamic model manipulation. These factors limit the platform's potential for a more immersive and intuitive learning experience. Furthermore, the user evaluation conducted so far is limited, with feedback gathered from a small group of participants. A more comprehensive evaluation involving a broader, diverse user base is necessary to assess the platform's usability and educational effectiveness fully. This broader evaluation would help determine how well the system meets the needs of a wide range of users and guide future improvements.

9 CONCLUSIONS AND FUTURE WORK

The 3D immersive interface offers unique advantages in anatomy education, transforming traditional learning methods through cutting-edge virtual reality. By providing a stereoscopic presentation of 3D visuals and fostering deep user immersion, the platform eliminates external distractions, enhances focus, and engages learners through gamified experiences. Its ability to deliver a multi-sensory, highly interactive environment significantly enhances comprehension and retention, with the emotional impact of immersive VR further amplifying the learning process.

This enhanced version of the immersive 3D Atlas of the Human Body introduces innovative learning opportunities, notably the integration of original anatomical slices within the 3D models, bridging the gap between cross-sectional imaging and volumetric anatomy. Furthermore, the platform incorporates artificial intelligence through a multilingual Large Language Model that provides detailed anatomical descriptions in French, English, and Spanish, with an optional voice feature to support accessibility and global engagement.

By uniting high-fidelity visuals, interactivity, and AI-enhanced learning tools, this platform establishes itself as a comprehensive, dynamic, and effective educational resource poised to revolutionize the teaching and understanding of human anatomy.

Future iterations of the application aim to extend its functionality into VR environments. A planned version will incorporate the Logitech MX Stylus and be designed for the Meta Quest family of headsets, further enriching the immersive learning experience. This comprehensive integration of features ensures the application meets the needs of diverse educational settings, making it a revolutionary tool for anatomy education.

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